Riding the Wave?

Who Starts, Follows and Gains from ML Research Trends

A Pre-registered Study of ICLR Submissions, 2017–2026:
43,480 Discovery and 11,099 Held-out Papers

[Your name]

[Department], [Institution]

[Venue / seminar] [Date]

Outline

1. Motivation and Research Questions
2. Data and Measurement
3. Methodological Contribution: Hindsight Bias in Topic Maps
4. Empirical Strategy
5. Results
6. Held-out Replication
7. Conclusion

Motivation: Research Fashions in Machine Learning

Share of ICLR submissions that ...	2017	2026
propose a new architecture	24%	8%
build on a foundation model	0.3%	54%
carry a branded title ("NAME: subtitle")	9%	41%

Table 1: Field-wide shifts in content and in style.

- ML research moves in waves, in *what* is studied and in *how* it is presented.
- Are these fashions? If so: what do they displace, who drives them, and are followers rewarded?

Research Questions and Pre-registered Hypotheses

Question	Key hypothesis	Predicted sign
RQ1 Rise: what does a fashion displace?	**H1d**: prior foundation-model (FM) adoption in a topic → share of new-architecture papers	−
RQ2 Following: who introduces new work?	**H2b**: newcomer paper → new-method paper	−
	H2c: newcomer paper → benchmark/evaluation paper	+
RQ3 Reward: are followers rewarded?	**H3a**: topic maturity (prior papers in topic) → acceptance	−

Table 2: Selected hypotheses. Ten hypotheses registered in total.

Our RQ2 prediction followed the fashion view: newcomers imitate, insiders innovate.

Related Work and Positioning

Two views of who drives new work

- *Fashion and herding*: ideas spread by imitation (Abrahamson, 1991; Bikhchandani et al., 1992) $\Rightarrow$ newcomers follow
- *Outsider innovation*: new entrants bring new ideas (Azoulay et al., 2019); tradition vs. innovation (Foster et al., 2015)
- Team size: large teams develop, small teams disrupt (Wu et al., 2019)

Meta-research on ML

- Benchmark concentration (Koch et al., 2021)
- Values encoded in ML papers (Birhane et al., 2022)

Measurement challenges we address

1. Survivorship: only published papers observed
2. Hindsight: topics defined on the full period
3. Forking paths (Gelman & Loken, 2014) $\Rightarrow$ pre-registration (Nosek et al., 2018)

Data and Measurement

Construct	Operationalisation	Validation
Corpus	43,480 discovery + 11,099 held-out ICLR submissions (released former confirmation split), 2017–2026, from OpenReview: accepted, rejected, withdrawn	Held-out: never analysed before its single registered run; a pre-registered out-of-sample replication, not a certified untouched holdout
Topic	SPECTER2 embeddings of title + abstract (Singh et al., 2023); $k = 100$ clusters; held-out papers placed on the frozen maps	Agreement with ICLR areas: NMI 0.30 (keyword baseline 0.27); robust to $k = 75, 150$
Contribution type	LLM with a fixed codebook: method, benchmark/evaluation, empirical, theory, other	Blind second pass: $\kappa = 0.93$ on both splits (held-out: 555 papers)
Newcomer paper	No author has an earlier ICLR submission *in the topic* (73% of papers)	Exact OpenReview author IDs
Acceptance	Final decision	Sealed in code until each split's single registered run (held-out: the earlier keyword grid's overlap is unknown)

Table 3: Constructs, measures and checks.

Hindsight Bias in Topic Maps

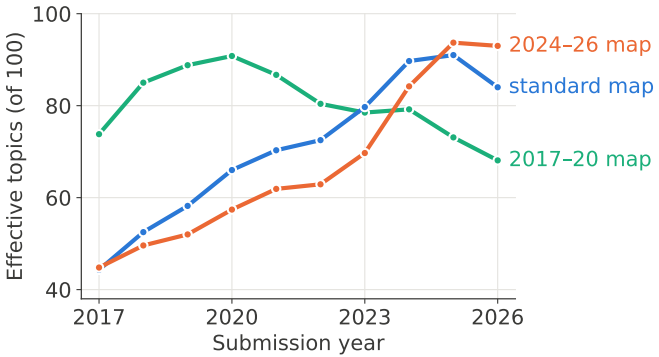


Figure 1: Effective number of topics per year under three $k = 100$ maps fitted on different years.

- 69% of submissions are from 2024–26: a map fitted on all years is a map of the *present*.
- Each map finds peak diversity in the years used to fit it.

Implication

Topic maps are not neutral preprocessing: the fitting window shapes the history one observes.

Consequence: Substantive Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** with the 2017–20 map vs. **29** with the standard map.
- Small topics 'predict' 2-year growth on the standard map ($R^2 = 0.28$), not on the 2017–20 map (-0.03). (Post hoc.)
- Do topics age, from new methods towards benchmarks? Same papers, same model:

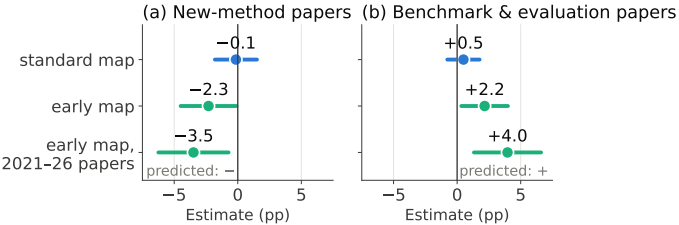


Figure 2: Coefficient on topic maturity (log prior papers), 95% CI. Early-map rows are exploratory.

⇒ Every registered result must also pass two hindsight checks (early-year and year-balanced maps).

Empirical Strategy

Specification (linear probability model)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{k(i)} \;+\; \tau_{t(i)} \;+\; \varepsilon_i$$

- $y_i \in \{0,1\}$: paper type or acceptance; x_i: e.g. newcomer paper
- μ_k, τ_t: topic and year fixed effects (topic $\times$ year for RQ2)
- Standard errors clustered by topic (100 clusters)

Decision rule (registered). Supported only if:

1. predicted sign, and Holm-adjusted $p < 0.05$;
2. same sign in $\geq 75\%$ of 9–11 alternative specifications;
3. same sign under both hindsight checks.

Protocol

Pre-registered; variables built blind to acceptance; 12,400 simulation runs; code and inputs frozen (SHA-256); a single run; every estimate reported.

Simulation finding

No-effect data: clustered SEs rejected up to 12.6% at nominal 5%, so RQ2 gets two extra checks (two-way clustering, a stress test).

Held-out rule (registered)

Registered tests only; one-sided $\alpha = 0.05$; frozen code and topic maps; features built on discovery + held-out papers.

RQ1: Foundation-Model Adoption and New Architectures

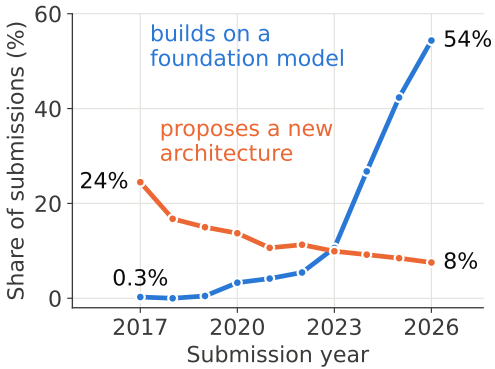


Figure 3: Field-wide trends (descriptive).

Within topic, net of year effects: prior FM adoption vs. new-architecture papers.

	Discovery	Held-out
$\hat{\beta}$ (0 $\rightarrow$ 100% FM)	-0.080	$\mathbf{-0.110}$
(SE)	(0.021)	(0.036)
95% CI	$[-0.121, -0.038]$	$[-0.181, -0.039]$
p	Holm 0.001	one-sided 0.001
Same sign (specs)	9 / 9	9 / 9
N	42,990	10,978

- H1d **supported**, and **replicated** on the held-out papers.
- Topics at 50% vs. 10% FM adoption: about one third fewer new-architecture papers (discovery; base 9%).

RQ2: Newcomers and Contribution Types

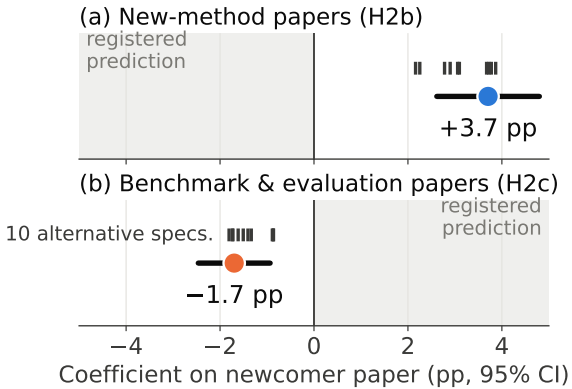


Figure 4: Discovery; topic × year FE; grey: predicted side.

	H2b (method)	H2c (bench.)
Discovery $\hat{\beta}$ (SE)	$+0.037$ (0.006)	-0.017 (0.004)
Same sign	10/10	10/10
Held-out $\hat{\beta}$	$\mathbf{+0.036}$	$\mathbf{-0.021}$
95% CI	$[0.015, 0.057]$	$[-0.033, -0.008]$
Base rate: disc.	75%	7.7%
held-out	75%	7.9%

- Discovery: both opposite to prediction (wild cluster bootstrap $p < 0.001$; $N = 42{,}315$)
- Held-out: reversed predictions H2b′/H2c′ (registered) **supported** (one-sided Holm $p = 0.001$ each; $N = 10{,}718$)
- Consistent with outsider innovation, not with herding

Most of the H2c gap goes with team size; H2b does not.

RQ3: Is Following the Fashion Rewarded?

Association with acceptance	Discovery	Held-out	Status
Topic maturity, per log unit of prior papers (H3a)	-1.7 pp (SE 1.0)	-1.5 pp (SE 1.9)	inconclusive in both
Branded title	ratio 1.02	not tested	exploratory
New arch. $\times$ prior FM adoption (0 $\rightarrow$ 100%)	$+8.3$ pp (SE 3.5)	not tested	exploratory, $p = 0.02$

Table 4: Topic and year fixed effects; SEs clustered by topic; decided papers, 2018–2026 (H3a: $N = 32{,}190$, MDE_{80} 4 pp; held-out $N = 8{,}185$, MDE_{80} 4.8 pp, one-sided $p = 0.21$).

Reading

No detectable reward for following the fashion at review; the crowding test is inconclusive in both splits. If anything, papers that go against the tide (architectures where FMs dominate) do relatively better (exploratory).

Far Jumps Go with Lower Acceptance (Post Hoc, Replicated)

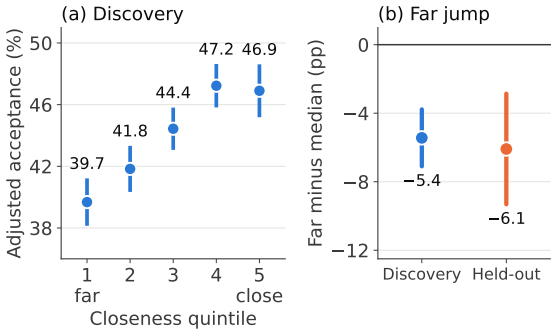


Figure 5: Decided papers with ≥ 1 experienced author, 2023–26 (E1 sample: discovery 20,430; held-out 5,344). Closeness: highest cosine between the abstract and an experienced co-author's earlier ICLR work. (b) 10th percentile minus median. 95% CIs.

- Far jump from the team's own record vs. median: -5.4 pp in discovery; -6.1 pp held-out (CI -9.3 to -2.9; Holm $p = 0.005$)
- Same author, closer paper: $+0.55$ pp (discovery), $+1.06$ pp (held-out) per $+0.1$ cosine
- Near vs. median: not lower ($+1.7$ pp discovery; $+2.3$ pp held-out, not registered)
- **Not replicated**: the field-novelty floor. Least-novel fifth: -3.3 pp in discovery, -0.1 pp held-out (CI -3.5 to $+3.2$); uninformative by the registered rule (power 0.47, MDE_{80} 5.4 pp)

Post hoc in discovery; held-out tests registered before the run. Held-out: post-decision abstracts, no cleaned-text check. Associations: co-author choice and failed projects not controlled.

Why Do Some Areas Lose Favour? (Post Hoc)

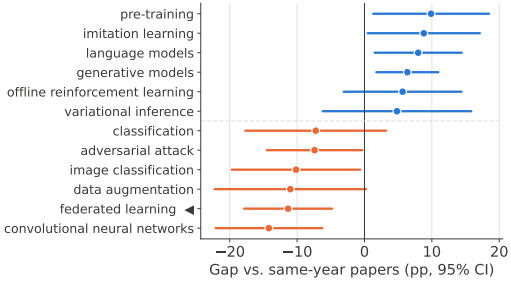


Figure 6: Acceptance gap vs. same-year papers, 2018–2026: six highest and six lowest author keywords ($\geq$5 years of data; 32,190 decided papers).

Pattern ($\pm$10 pp): current-wave and specialist areas above; older or entry-level areas below.

Candidate explanations (checked across 50 keywords):

- *Entry points attract newcomers:* 85–88% of CNN and classification papers come from authors new to the topic; more newcomers, lower acceptance ($\rho = -0.40$, $p = 0.004$).
- *Paradigm moved on:* CNN and image-classification shares peaked in 2018 (2026: 5% and 24% of peak); incremental gains on saturated benchmarks may face a higher bar (untested).
- *Cooling area:* federated learning, with established authors and a falling share since 2023.
- *Not* paper type, team size (adjusted vs. raw gaps $r = 0.97$) *or* the FM wave (FM share vs. acceptance $\rho = 0.07$).

Associations, not causes; no area survives correction for 68 tests. Held-out: edges shrink ($0.60\times$) but persist; FL gap not confirmed.

Do the Results Hold on Held-out Papers?

Test	Pred.	Discovery	Held-out [95% CI]	One-sided p	Verdict
Registered hypotheses (one-sided p, Holm-adjusted within family)					
H1d FM adoption ($0 \to 1$) $\to$ new arch.	$-$	-8.0	-11.0 [-18.1, -3.9]	0.001	supported
H2b′ newcomer $\to$ method paper	$+$	$+3.7$	$+3.6$ [$+1.5$, $+5.7$]	0.001	supported
H2c′ newcomer $\to$ benchmark paper	$-$	-1.7	-2.1 [-3.3, -0.8]	0.001	supported
H3a maturity (log) $\to$ acceptance	$-$	-1.7	-1.5 [-5.2, $+2.2$]	0.21	inconclusive†
H4-FL FL keyword $\to$ acceptance	$-$	-11.3	-4.9 [-15.4, $+5.7$]	0.14	not supported†
PH5 primary (post hoc in discovery; one-sided p, Holm-adjusted over the pair)					
E1 far jump (p10 $-$ median)	$-$	-5.4	-6.1 [-9.3, -2.9]	0.005	supported
P1b least-novel $-$ middle fifth	$-$	-3.3	-0.1 [-3.5, $+3.2$]	0.47	not supported†
PH5 secondary (one-sided p, unadjusted)					
E4 within author, per $+0.1$ cos	$+$	$+0.55$	$+1.06$ [$+0.23$, $+1.89$]	0.006	supported
Teams H1 one insider vs none	$+$	$+3.4$	$+3.9$ [$+0.3$, $+7.5$]	0.017	supported
A1 Emerging $-$ Mainstream keyword	$+$	$+2.7$	$+3.3$ [-0.2, $+6.7$]	0.031	supported
Premia b topic edges (early map)	$0 < b < 1$	0.38‡	0.60 [0.32, 0.88]	0.003	supported

Table 5: 11,099 released former confirmation papers this study had never analysed: a pre-registered out-of-sample replication, not a certified untouched holdout. Supported: one-sided $p < 0.05$, Holm within families (so a 95% CI can cross 0). pp unless noted. † Below MDE_{80}: uninformative. ‡ Within-discovery slope. Also registered: H4-FL trend, P7 not supported; premia EB supported.

Contributions

1. **Methodological.** Topic maps fitted with hindsight manufacture emerging topics and can reverse trend conclusions; early-year maps provide a cheap check.
2. **Design.** A pre-registered, outcome-blind, simulation-checked, single-run protocol for science-of-science studies, with a registered held-out replication.
3. **Substantive** (associations).
   - **Replicated:** where foundation models spread, new architectures fade (H1d); newcomer papers lean towards new methods and away from benchmarks (H2b′/H2c′); far jumps from the team's own record go with lower acceptance (post hoc in discovery).
   - **Not replicated:** the field-novelty floor (least-novel fifth); the held-out test was uninformative (power 0.47, MDE_{80} 5.4 pp).
   - **Not confirmed:** the federated-learning acceptance gap; the held-out test was uninformative (134 papers, MDE_{80} 14.5 pp).
   - **Inconclusive** in both splits: crowded (mature) topics and acceptance.

Limitations and Future Work

Limitations

- One venue (ICLR)
- Associations, not causal effects
- LLM labels: consistent ($\kappa = 0.93$), not yet human-validated
- Camera-ready text: the discovery cleaned-text checks held; no such check for held-out papers
- Held-out papers are a released former holdout, not certified untouched
- Pooled re-run (54,579 papers) is exploratory and confirms nothing
- 2021 author-ID change undercounts incumbents (expected bias towards zero)

Future work

- A prospective test on future ICLR submissions
- Extension to ACL Anthology / ARR
- Human validation of contribution labels
- Review scores: reviewers vs. final decisions

Thank you

Questions and comments welcome

[email]

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